

Laboratorio: Realización de una Búsqueda Condicional (DQL Avanzado)

Atributo	Detalle
Dificultad	Intermedio
Tiempo Estimado	45 minutos
Servicios Principales	Amazon EC2 (Command Host), MySQL Engine

Resumen y Objetivos

Este laboratorio profundiza en el uso de la cláusula `WHERE` para realizar búsquedas granulares y precisas dentro de un conjunto de datos. Aprenderás a utilizar operadores lógicos y funciones integradas para filtrar registros basándote en rangos, patrones de texto y cálculos agregados.

Al finalizar este laboratorio, serás capaz de:

- 1. Escribir condiciones de búsqueda complejas utilizando la cláusula `WHERE`.
 - 2. Implementar el operador `BETWEEN` para filtrado de rangos inclusivos.
 - 3. Utilizar el operador `LIKE` con caracteres comodín (`%`) para búsqueda de patrones.
 - 4. Aplicar alias de columna con `AS` y funciones de agregación como `SUM()`.
 - 5. Manejar la sensibilidad a mayúsculas y minúsculas mediante la función `LOWER()`.
-

Análisis del Escenario

Como parte del equipo de operaciones de base de datos, se te ha solicitado extraer métricas específicas de la tabla `country`. El diagnóstico inicial indica que las consultas simples ya no son suficientes; ahora se requiere consolidar datos demográficos por regiones geográficas y filtrar países dentro de umbrales poblacionales específicos. La precisión en el uso de operadores lógicos es crítica para evitar resultados erróneos o "ruido" en los reportes.

Arquitectura

La estructura de trabajo se compone de:

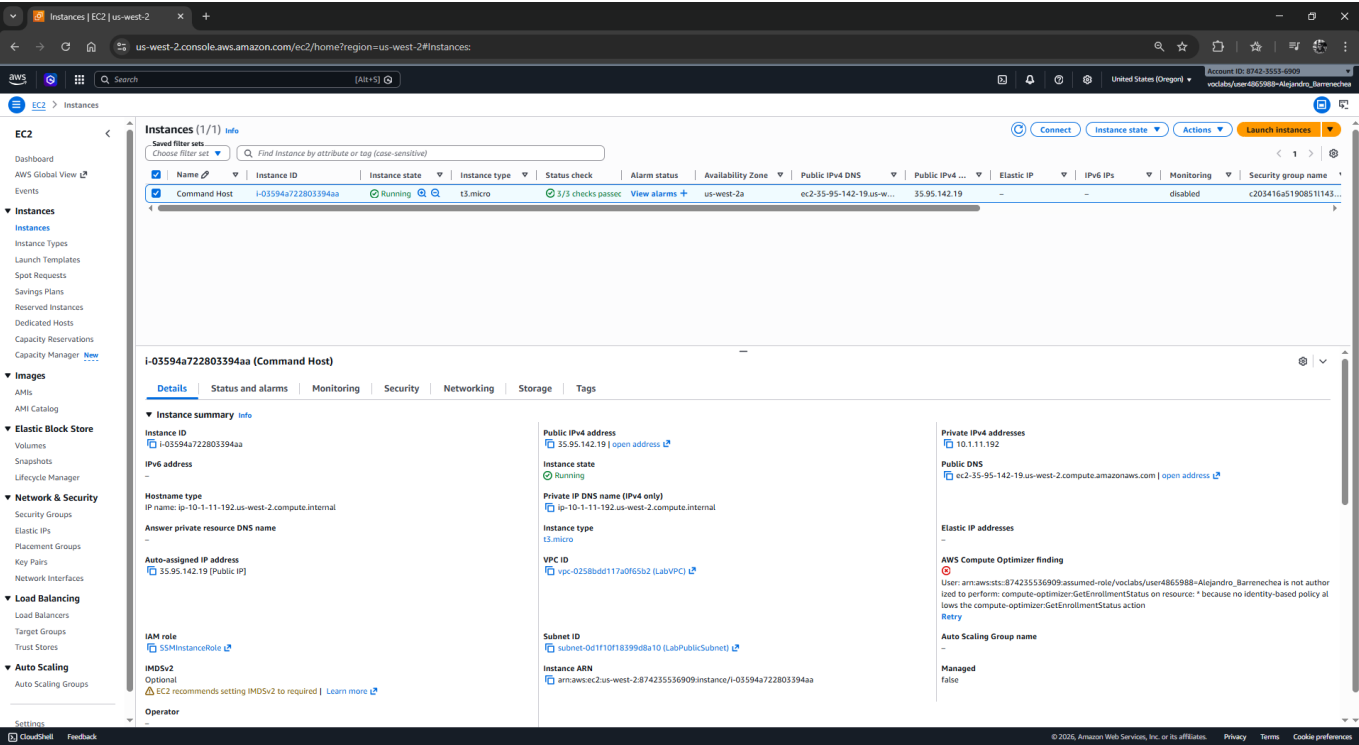
- **Command Host (EC2):** Tu estación de trabajo Linux con el cliente `mysql` instalado.
 - **Motor de Base de Datos:** Instancia local de MySQL que aloja la base de datos `world`.
 - **Lógica de Consulta:** Interacción mediante SQL Shell para procesar funciones de agregación y filtros condicionales.
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Desarrollo de las Tareas Paso a Paso

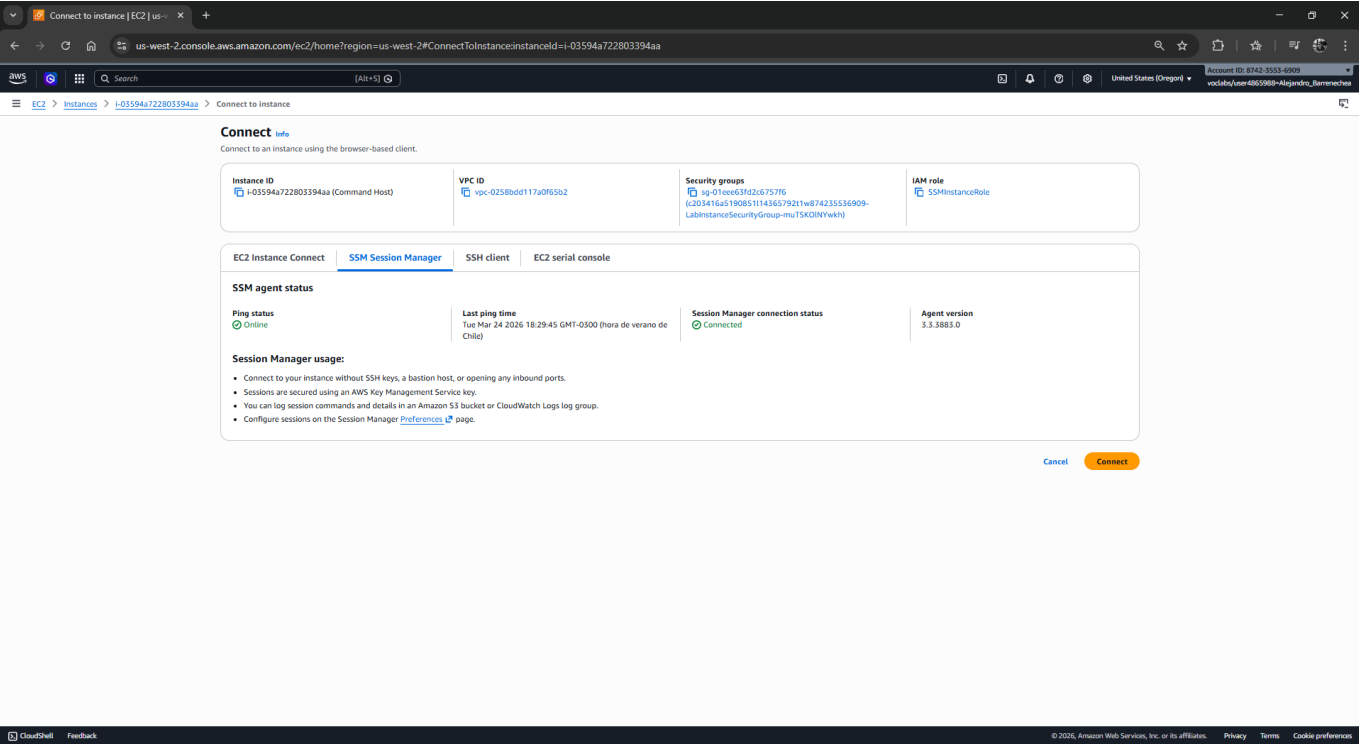
Tarea 1: Conexión al Command Host

Establece la conexión segura con la infraestructura de base de datos.

- 1. Ingresa a la consola de **Amazon EC2**.
- 2. En el menú de navegación, haz clic en **Instances** (Instancias).
- 3. Ubica el **Command Host**, selecciónalo y pulsa en **Connect**.



- 4. Selecciona la pestaña **Session Manager** y haz clic en el botón naranja **Connect**.

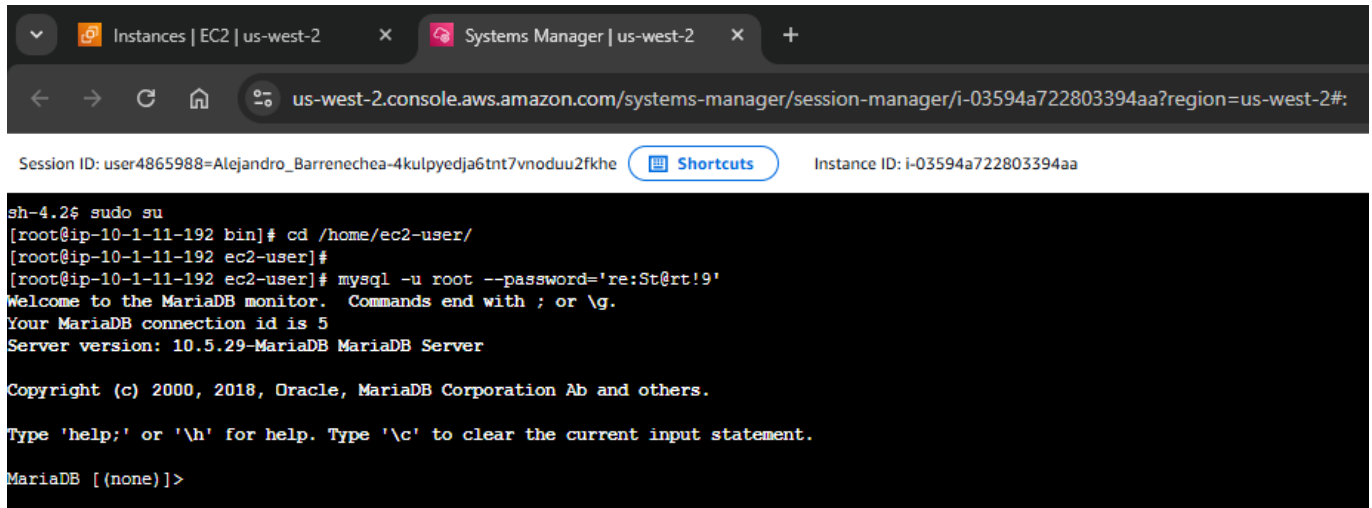


- 5. Prepara el shell ejecutando:

```
sudo su  
cd /home/ec2-user/
```

6. Inicia la sesión en MySQL:

```
mysql -u root --password='re:St@rt!9'
```



The screenshot shows the AWS Systems Manager console interface. At the top, there are tabs for 'Instances | EC2 | us-west-2' and 'Systems Manager | us-west-2'. The browser address bar shows the URL: `us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-03594a722803394aa?region=us-west-2#`. Below the address bar, the session ID is `user4865988=Alejandro_Barrenechea-4kulpyedja6tnt7vnoduu2fkhe` and the instance ID is `i-03594a722803394aa`. The main terminal window displays the following commands and output:

```
sh-4.2$ sudo su  
[root@ip-10-1-11-192 bin]# cd /home/ec2-user/  
[root@ip-10-1-11-192 ec2-user]#  
[root@ip-10-1-11-192 ec2-user]# mysql -u root --password='re:St@rt!9'  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 5  
Server version: 10.5.29-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]>
```

Tarea 2: Consulta de la Base de Datos `world`

Ejecutarás filtros avanzados para extraer información de valor.

1. Verifica la existencia de la base de datos: `SHOW DATABASES;`.
2. Inspecciona los datos generales: `SELECT * FROM world.country;`.

```

Session ID: user4865988-Alejandro_Barrenechea-4kulpdydja6t7vmodu2fhe Instance ID: I-03594a722803394aa
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| world |
+-----+
8 rows in set (0.001 sec)

MariaDB [(none)]> SELECT * FROM world.country;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Code | Name | HeadOfState | Continent | Region | SurfaceArea | IndepYear | Population | LifeExpectancy | GNP | GNPOld | LocalName | GovernmentForm |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| ASA | Aruba | | North America | Caribbean | 193.00 | NULL | 103000 | 78.4 | 825.00 | 793.00 | Aruba | Nonmetropolitan Territory of The Netherla |
| BSA | Beatrix | 129 | AM | | 129 | 1919 | 22720000 | 45.9 | 5976.00 | NULL | Afghanistan/Afghanistan | Islamic Emirate |
| AFG | Afghanistan | | Asia | Southern and Central Asia | 652090.00 | 1919 | 22720000 | 45.9 | 5976.00 | NULL | Afghanistan/Afghanistan | Islamic Emirate |
| MOH | Mohamed Omar | 1 | AF | | 1246700.00 | 1975 | 12878000 | 38.3 | 6648.00 | 7984.00 | Angola | Republic |
| AGO | Angola | | Africa | Central Africa | 1246700.00 | 1975 | 12878000 | 38.3 | 6648.00 | 7984.00 | Angola | Republic |
| JES | Jos   Eduardo dos Santos | 56 | AO | | 96.00 | NULL | 8000 | 76.1 | 63.20 | NULL | Anguilla | Dependent Territory of the UK |
| AIA | Anguilla | 62 | AI | | 28748.00 | 1912 | 3401200 | 71.6 | 3205.00 | 2500.00 | Shqip ria | Republic |
| ALB | Albania | | Europe | Southern Europe | 28748.00 | 1912 | 3401200 | 71.6 | 3205.00 | 2500.00 | Shqip ria | Republic |
| RSD | Rados r Wujdani | 34 | RS | | 468.00 | 1278 | 78000 | 83.5 | 1630.00 | NULL | Andorra | Parliamentary Coprincipality |
| AND | Andorra | | Europe | Southern Europe | 468.00 | 1278 | 78000 | 83.5 | 1630.00 | NULL | Andorra | Parliamentary Coprincipality |
| ANT | Netherlands Antilles | 55 | AD | | 800.00 | NULL | 217000 | 74.7 | 1941.00 | NULL | Nederlandse Antillen | Nonmetropolitan Territory of The Netherla |
| BSA | Beatrix | 33 | AN | | 83600.00 | 1971 | 2441000 | 74.1 | 37966.00 | 36846.00 | Al-Imarat al-Arabiya al-Muttahida | Emirate Federation |
| ARE | United Arab Emirates | | Asia | Middle East | 83600.00 | 1971 | 2441000 | 74.1 | 37966.00 | 36846.00 | Al-Imarat al-Arabiya al-Muttahida | Emirate Federation |
| EGY | Egypt | 25 | EG | | 2780400.00 | 1816 | 37032000 | 75.1 | 340238.00 | 323310.00 | Argentina | Federal Republic |
| ARG | Argentina | | South America | South America | 2780400.00 | 1816 | 37032000 | 75.1 | 340238.00 | 323310.00 | Argentina | Federal Republic |
| FER | Fernando de la R  a | 69 | AR | | 29800.00 | 1991 | 3520000 | 66.4 | 1813.00 | 1627.00 | Hajastan | Republic |
| ARM | Armenia | 126 | AM | | 199.00 | NULL | 68000 | 75.1 | 334.00 | NULL | Amerika Samoa | US Territory |
| ASM | American Samoa | | Oceania | Polynesia | 199.00 | NULL | 68000 | 75.1 | 334.00 | NULL | Amerika Samoa | US Territory |
| GEA | George H. Bush | 24 | AS | | 13120000.00 | NULL | 0 | NULL | 0.00 | NULL |     | Co-administrated |
| ATA | Antarctica | | Antarctica | Antarctica | 13120000.00 | NULL | 0 | NULL | 0.00 | NULL |     | Co-administrated |
| FRA | French Southern territories | NULL | AQ | | 7780.00 | NULL | 0 | NULL | 0.00 | NULL | Terres australes fran aises | Nonmetropolitan Territory of France |
| JCS | Jacques Chirac | NULL | TF | | 442.00 | 1981 | 68000 | 70.5 | 612.00 | 584.00 | Antigua and Barbuda | Constitutional Monarchy |
| ATG | Antigua and Barbuda | | North America | Caribbean | 442.00 | 1981 | 68000 | 70.5 | 612.00 | 584.00 | Antigua and Barbuda | Constitutional Monarchy |
| ELS | Elisabeth II | 63 | AG | | 7741220.00 | 1901 | 18886000 | 79.8 | 351182.00 | 392911.00 | Australia | Constitutional Monarchy, Federation |
| AUS | Australia | | Oceania | Australia and New Zealand | 7741220.00 | 1901 | 18886000 | 79.8 | 351182.00 | 392911.00 | Australia | Constitutional Monarchy, Federation |
| ELS | Elisabeth II | 135 | AU | |

```

```

Session ID: user4865988-Alejandro_Barrenechea-4kulpdydja6t7vmodu2fhe Instance ID: I-03594a722803394aa
| JOA | Jos   Alexandre Guzm  n | 1522 | TP | | Polynesia | 650.00 | 1970 | 99000 | 67.9 | 146.00 | 170.00 | Tonga | Monarchy |
| TON | Tonga | | Oceania | Polynesia | 650.00 | 1970 | 99000 | 67.9 | 146.00 | 170.00 | Tonga | Monarchy |
| TAU | Taufa'au Tupou IV | 3334 | TO | | 5130.00 | 1962 | 1295000 | 68.0 | 6232.00 | 5867.00 | Trinidad and Tobago | Republic |
| TTO | Trinidad and Tobago | | North America | Caribbean | 5130.00 | 1962 | 1295000 | 68.0 | 6232.00 | 5867.00 | Trinidad and Tobago | Republic |
| ARB | Arthur M. B. Robinson | 3336 | TT | | 163610.00 | 1956 | 9586000 | 73.7 | 20026.00 | 18898.00 | T  nis/Tunisie | Republic |
| TUN | Tunisia | | Africa | Northern Africa | 163610.00 | 1956 | 9586000 | 73.7 | 20026.00 | 18898.00 | T  nis/Tunisie | Republic |
| ZIN | Zine el-Abidine Ben Ali | 3349 | TN | | 774815.00 | 1923 | 66591000 | 71.0 | 210721.00 | 159122.00 | T  nkiye | Republic |
| TUR | Turkey | | Asia | Middle East | 774815.00 | 1923 | 66591000 | 71.0 | 210721.00 | 159122.00 | T  nkiye | Republic |
| AHE | Ahmet Necdet Sezer | 3358 | TR | | 26.00 | 1978 | 12000 | 66.3 | 6.00 | NULL | Tuvalu | Constitutional Monarchy |
| TUV | Tuvalu | | Oceania | Polynesia | 26.00 | 1978 | 12000 | 66.3 | 6.00 | NULL | Tuvalu | Constitutional Monarchy |
| ELS | Elisabeth II | 3424 | TV | | 36188.00 | 1945 | 22256000 | 76.4 | 256254.00 | 263451.00 | T  i-wan | Republic |
| TWN | Taiwan | | Asia | Eastern Asia | 36188.00 | 1945 | 22256000 | 76.4 | 256254.00 | 263451.00 | T  i-wan | Republic |
| CHS | Chen Shui-bian | 3263 | TW | | 883749.00 | 1961 | 33517000 | 52.3 | 8005.00 | 7388.00 | Tanzania | Republic |
| TZA | Tanzania | | Africa | Eastern Africa | 883749.00 | 1961 | 33517000 | 52.3 | 8005.00 | 7388.00 | Tanzania | Republic |
| BEN | Benjamin William Mkapa | 3306 | TZ | | 241038.00 | 1962 | 21778000 | 42.9 | 6313.00 | 6887.00 | Uganda | Republic |
| UGA | Uganda | | Africa | Eastern Africa | 241038.00 | 1962 | 21778000 | 42.9 | 6313.00 | 6887.00 | Uganda | Republic |
| YOW | Yoweri Museveni | 3425 | UG | | 603700.00 | 1991 | 50456000 | 66.0 | 42168.00 | 49677.00 | Ukr  ina | Republic |
| UKR | Ukraine | | Europe | Eastern Europe | 603700.00 | 1991 | 50456000 | 66.0 | 42168.00 | 49677.00 | Ukr  ina | Republic |
| ECU | Ecuador | 3426 | EA | | 16.00 | NULL | 0 | NULL | 0.00 | NULL | United States Minor Outlying Islands | Dependent Territory of the US |
| UMI | United States Minor Outlying Islands | | Oceania | Micronesia/Caribbean | 16.00 | NULL | 0 | NULL | 0.00 | NULL | United States Minor Outlying Islands | Dependent Territory of the US |
| GEA | George H. Bush | NULL | UM | | 175016.00 | 1828 | 3337000 | 75.2 | 20831.00 | 19967.00 | Uruguay | Republic |
| URU | Uruguay | | South America | South America | 175016.00 | 1828 | 3337000 | 75.2 | 20831.00 | 19967.00 | Uruguay | Republic |
| JOR | Jorge Batlle Iba  ez | 3492 | UY | | 9363520.00 | 1776 | 278357000 | 77.1 | 8510700.00 | 8110900.00 | United States | Federal Republic |
| USA | United States | | North America | North America | 9363520.00 | 1776 | 278357000 | 77.1 | 8510700.00 | 8110900.00 | United States | Federal Republic |
| GEA | George H. Bush | 3113 | US | | 447400.00 | 1991 | 24318000 | 63.7 | 14194.00 | 21300.00 | Uzbekistan | Republic |
| UZB | Uzbekistan | | Asia | Southern and Central Asia | 447400.00 | 1991 | 24318000 | 63.7 | 14194.00 | 21300.00 | Uzbekistan | Republic |
| ISL | Islam Karimov | 3503 | UZ | | 10.40 | 1929 | 1000 | NULL | 9.00 | NULL | Santa Sede/Citt   del Vaticano | Independent Church State |
| VAT | Holy See (Vatican City State) | | Europe | Southern Europe | 10.40 | 1929 | 1000 | NULL | 9.00 | NULL | Santa Sede/Citt   del Vaticano | Independent Church State |
| JOH | Johannes Pavelli II | 3538 | VA | | 388.00 | 1979 | 114000 | 72.3 | 285.00 | NULL | Saint Vincent and the Grenadines | Constitutional Monarchy |
| VCT | Saint Vincent and the Grenadines | | North America | Caribbean | 388.00 | 1979 | 114000 | 72.3 | 285.00 | NULL | Saint Vincent and the Grenadines | Constitutional Monarchy |
| ELS | Elisabeth II | 3066 | VC | | 912050.00 | 1811 | 24170000 | 73.1 | 95023.00 | 88434.00 | Venezuela | Federal Republic |
| VEN | Venezuela | | South America | South America | 912050.00 | 1811 | 24170000 | 73.1 | 95023.00 | 88434.00 | Venezuela | Federal Republic |
| HUG | Hugo Ch  vez Fr  s | 3539 | VE | | 151.00 | NULL | 21000 | 75.4 | 612.00 | 573.00 | British Virgin Islands | Dependent Territory of the UK |
| VIR | Virgin Islands, British | | North America | Caribbean | 151.00 | NULL | 21000 | 75.4 | 612.00 | 573.00 | British Virgin Islands | Dependent Territory of the UK |
| VIR | Virgin Islands, U.S. | | North America | Caribbean | 347.00 | NULL | 93000 | 78.1 | 0.00 | NULL | Virgin Islands of the United States | US Territory |
| GEA | George H. Bush | 4567 | VI | | 331689.00 | 1945 | 79832000 | 69.3 | 21929.00 | 22834.00 | VLA* Nam | Socialist Republic |
| VNM | Vietnam | | Asia | Southeast Asia | 331689.00 | 1945 | 79832000 | 69.3 | 21929.00 | 22834.00 | VLA* Nam | Socialist Republic |
| TRN | Tr  n Duc Lu  ng | 3770 | VN | | 12189.00 | 1980 | 190000 | 60.6 | 261.00 | 246.00 | Vanuatu | Republic |
| VUT | Vanuatu | | Oceania | Melanesia | 12189.00 | 1980 | 190000 | 60.6 | 261.00 | 246.00 | Vanuatu | Republic |
| JON | John Bani | 3537 | VU | | 200.00 | NULL | 15000 | NULL | 0.00 | NULL | Wallis-et-Futuna | Nonmetropolitan Territory of France |
| WLF | Wallis et Futuna | | Oceania | Polynesia | 200.00 | NULL | 15000 | NULL | 0.00 | NULL | Wallis-et-Futuna | Nonmetropolitan Territory of France |
| JCS | Jacques Chirac | 3536 | WF | | 2831.00 | 1962 | 180000 | 69.2 | 141.00 | 157.00 | Samoa | Parliamentary Monarchy |
| WSM | Samoa | | Oceania | Polynesia | 2831.00 | 1962 | 180000 | 69.2 | 141.00 | 157.00 | Samoa | Parliamentary Monarchy |
| MAL | Mawlaica Tammamfilli II | 3169 | MS | | 527968.00 | 1918 | 18112000 | 59.8 | 6041.00 | 5728.00 | Al-Yaman | Republic |
| YEM | Yemen | | Asia | Middle East | 527968.00 | 1918 | 18112000 | 59.8 | 6041.00 | 5728.00 | Al-Yaman | Republic |
| ALI | Abdallah Salih | 1780 | YE | | 102173.00 | 1918 | 10640000 | 72.4 | 17000.00 | NULL | Jugoslavia | Federal Republic |
| YUG | Yugoslavia | | Europe | Southern Europe | 102173.00 | 1918 | 10640000 | 72.4 | 17000.00 | NULL | Jugoslavia | Federal Republic |
| VOJ | Vojislav Koltumica | 1752 | YU | | 1221037.00 | 1910 | 40377000 | 51.1 | 116729.00 | 129092.00 | South Africa | Republic |
| ZAF | South Africa | | Africa | Southern Africa | 1221037.00 | 1910 | 40377000 | 51.1 | 116729.00 | 129092.00 | South Africa | Republic |
| THO | Thabo Mbeki | 716 | ZA | | 752618.00 | 1964 | 9169000 | 37.2 | 3377.00 | 3922.00 | Zambia | Republic |
| ZMB | Zambia | | Africa | Eastern Africa | 752618.00 | 1964 | 9169000 | 37.2 | 3377.00 | 3922.00 | Zambia | Republic |
| FRI | Frederick Chiluba | 3162 | ZM | | 390757.00 | 1980 | 11669000 | 37.8 | 5951.00 | 8670.00 | Zimbabwe | Republic |
| ZIM | Zimbabwe | | Africa | Eastern Africa | 390757.00 | 1980 | 11669000 | 37.8 | 5951.00 | 8670.00 | Zimbabwe | Republic |
| ROBERT | Robert G. Mugabe | 4068 | ZW | |
239 rows in set (0.001 sec)

MariaDB [(none)]>

```

Filtrado por Rangos (AND vs BETWEEN)

3. Filtra pa  ses con una poblaci  n entre 50 y 100 millones usando operadores de comparaci  n:

```

SELECT Name, Capital, Region, SurfaceArea, Population
FROM world.country
WHERE Population >= 50000000 AND Population <= 100000000;

```

4. Optimiza la legibilidad de la consulta anterior utilizando el operador **BETWEEN**:

```
SELECT Name, Capital, Region, SurfaceArea, Population
FROM world.country
WHERE Population BETWEEN 50000000 AND 100000000;
```

```

MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population
-> FROM world.country
-> WHERE Population >= 50000000 AND Population <= 100000000;
+-----+-----+-----+-----+-----+
| Name | Capital | Region | SurfaceArea | Population |
+-----+-----+-----+-----+-----+
| Congo, The Democratic Republic of the | 2298 | Central Africa | 2344858.00 | 51654000 |
| Germany | 3068 | Western Europe | 357022.00 | 82164700 |
| Egypt | 608 | Northern Africa | 1001449.00 | 88470000 |
| Ethiopia | 756 | Eastern Africa | 1104300.00 | 62865000 |
| France | 2974 | Western Europe | 551500.00 | 59225700 |
| United Kingdom | 486 | British Islands | 242900.00 | 59623400 |
| Iran | 1380 | Southern and Central Asia | 1648199.00 | 67702000 |
| Italy | 1464 | Southern Europe | 301316.00 | 57680000 |
| Mexico | 2515 | Central America | 1958201.00 | 98881000 |
| Philippines | 766 | Southeast Asia | 300000.00 | 75967000 |
| Thailand | 3320 | Southeast Asia | 513115.00 | 61399000 |
| Turkey | 3388 | Middle East | 774815.00 | 68591000 |
| Ukraine | 3426 | Eastern Europe | 603700.00 | 50456000 |
| Vietnam | 3770 | Southeast Asia | 331689.00 | 79832000 |
+-----+-----+-----+-----+-----+
14 rows in set (0.000 sec)

MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population
-> FROM world.country
-> WHERE Population BETWEEN 50000000 AND 100000000;
+-----+-----+-----+-----+-----+
| Name | Capital | Region | SurfaceArea | Population |
+-----+-----+-----+-----+-----+
| Congo, The Democratic Republic of the | 2298 | Central Africa | 2344858.00 | 51654000 |
| Germany | 3068 | Western Europe | 357022.00 | 82164700 |
| Egypt | 608 | Northern Africa | 1001449.00 | 88470000 |
| Ethiopia | 756 | Eastern Africa | 1104300.00 | 62865000 |
| France | 2974 | Western Europe | 551500.00 | 59225700 |
| United Kingdom | 486 | British Islands | 242900.00 | 59623400 |
| Iran | 1380 | Southern and Central Asia | 1648199.00 | 67702000 |
| Italy | 1464 | Southern Europe | 301316.00 | 57680000 |
| Mexico | 2515 | Central America | 1958201.00 | 98881000 |
| Philippines | 766 | Southeast Asia | 300000.00 | 75967000 |
| Thailand | 3320 | Southeast Asia | 513115.00 | 61399000 |
| Turkey | 3388 | Middle East | 774815.00 | 68591000 |
| Ukraine | 3426 | Eastern Europe | 603700.00 | 50456000 |
| Vietnam | 3770 | Southeast Asia | 331689.00 | 79832000 |
+-----+-----+-----+-----+-----+
14 rows in set (0.000 sec)

MariaDB [(none)]>

```

Búsqueda de Patrones y Agregación

- Calcula la población total de todas las regiones que contengan la palabra "Europe" utilizando el comodín %:

```
SELECT sum(Population) FROM world.country WHERE Region LIKE "%Europe%";
```

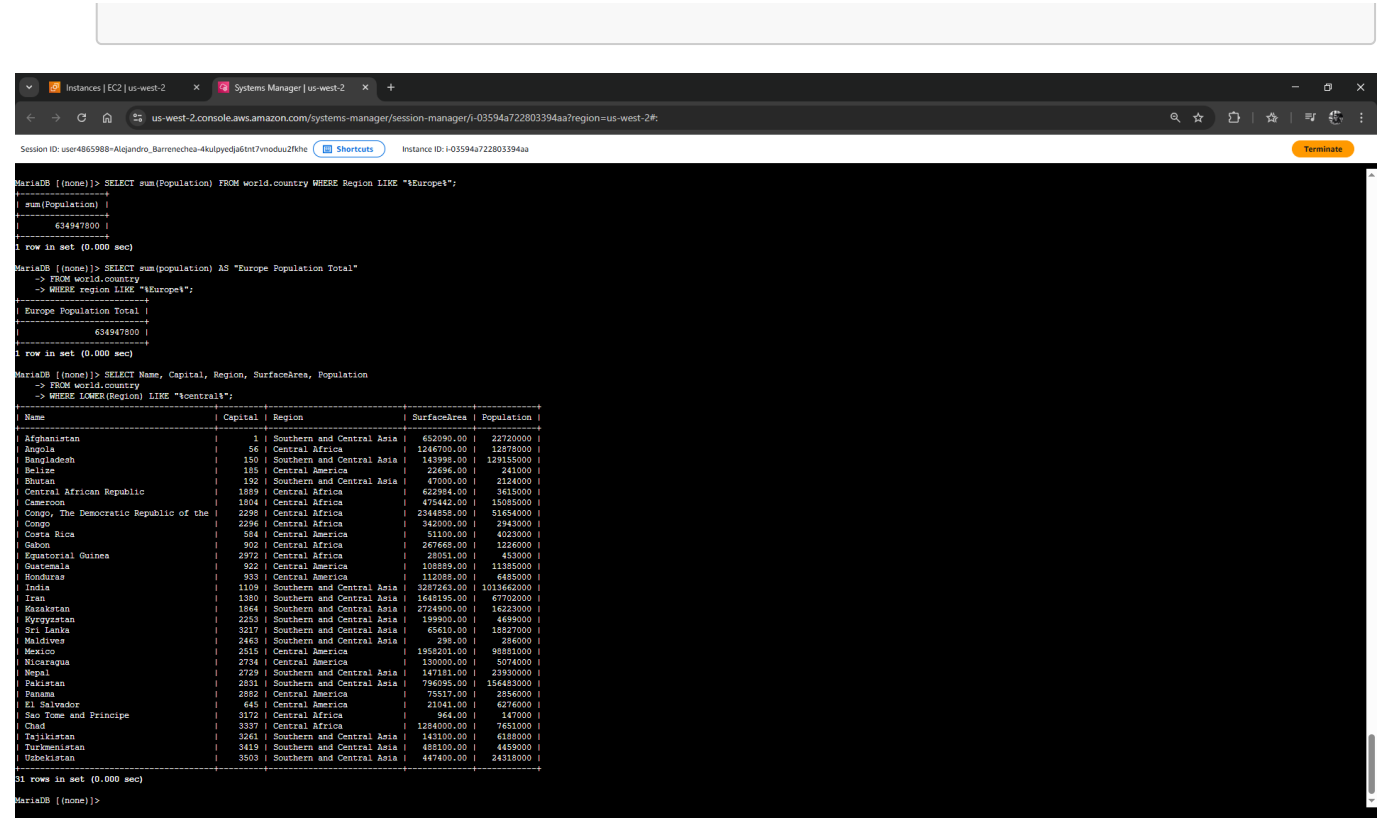
- Mejora la presentación del resultado anterior asignando un **Alias** a la columna calculada:

```
SELECT sum(population) AS "Europe Population Total"
FROM world.country
WHERE region LIKE "%Europe%";
```

Normalización de Texto (LOWER)

- Realiza una búsqueda que ignore la capitalización (case-insensitive) forzando el campo a minúsculas para comparar con la cadena "central":

```
SELECT Name, Capital, Region, SurfaceArea, Population
FROM world.country
WHERE LOWER(Region) LIKE "%central%";
```



Challenge (Desafío Técnico)

Instrucción: Escribe una consulta que devuelva la suma de la superficie (**SurfaceArea**) y la suma de la población (**Population**) de "North America".

Solución Sugerida:

```
SELECT sum(SurfaceArea) AS "Total Surface Area", sum(Population) AS "Total
Population"
FROM world.country
WHERE Continent = 'North America';
```

*Nota: Antes de ejecutar, asegúrate de validar si el filtro debe aplicarse a la columna **Continent** o **Region** según los datos observados en la Tarea 2.*

```
Instances | EC2 | us-west-2 x Systems Manager | us-west-2 x +
us-west-2.console.aws.amazon.com/systems-manager/instance-manager/i-03594a722803394aa?region=us-west-2#
Session ID: user4865988-Alejandro_Barrenechea-4kulpdyedj6t7mduau2fhe Shortcuts Instance ID: i-03594a722803394aa Terminate

-> FROM world.country
-> WHERE region LIKE "%Europe%";
-----+-----+-----+-----+-----+
| Europe Population Total |
|-----+-----+-----+-----+
| 634947800 |
+-----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population
-> FROM world.country
-> WHERE LOWER(region) LIKE "%central%";
-----+-----+-----+-----+-----+
| Name | Capital | Region | SurfaceArea | Population |
+-----+-----+-----+-----+-----+
| Afghanistan | 1 | Southern and Central Asia | 652090.00 | 22720000 |
| Angola | 56 | Central Africa | 1246700.00 | 12873000 |
| Bangladesh | 150 | Southern and Central Asia | 143998.00 | 129155000 |
| Belize | 185 | Central America | 22696.00 | 241000 |
| Bhutan | 192 | Southern and Central Asia | 37900.00 | 2124500 |
| Central African Republic | 1889 | Central Africa | 622984.00 | 3615000 |
| Cameroon | 1804 | Central Africa | 475442.00 | 15069000 |
| Congo, The Democratic Republic of the | 2388 | Central Africa | 2344859.00 | 51654000 |
| Congo | 2296 | Central Africa | 342000.00 | 2943000 |
| Costa Rica | 384 | Central America | 51100.00 | 4023000 |
| Gabon | 902 | Central Africa | 267468.00 | 1226000 |
| Equatorial Guinea | 2972 | Central Africa | 28051.00 | 453000 |
| Guatemala | 922 | Central America | 108889.00 | 11385000 |
| Honduras | 913 | Central America | 112089.00 | 4485000 |
| India | 1109 | Southern and Central Asia | 3287263.00 | 1013662000 |
| Iran | 1380 | Southern and Central Asia | 1648136.00 | 67702000 |
| Kazakhstan | 1844 | Southern and Central Asia | 2724900.00 | 15223000 |
| Kyrgyzstan | 2253 | Southern and Central Asia | 199900.00 | 4699000 |
| Sri Lanka | 3217 | Southern and Central Asia | 65410.00 | 1827000 |
| Maldives | 2463 | Southern and Central Asia | 299.00 | 284000 |
| Mexico | 2515 | Central America | 1958201.00 | 98881000 |
| Nicaragua | 2724 | Central America | 130000.00 | 5074000 |
| Nepal | 2729 | Southern and Central Asia | 147181.00 | 23930000 |
| Pakistan | 2831 | Southern and Central Asia | 796095.00 | 156483000 |
| Panama | 2882 | Central America | 75517.00 | 2856000 |
| El Salvador | 645 | Central America | 21041.00 | 6276000 |
| Sao Tome and Principe | 3172 | Central Africa | 964.00 | 147000 |
| Chad | 3337 | Central Africa | 1284000.00 | 7651000 |
| Tajikistan | 3261 | Southern and Central Asia | 143100.00 | 6188000 |
| Turkmenistan | 3419 | Southern and Central Asia | 488100.00 | 4459000 |
| Uzbekistan | 3503 | Southern and Central Asia | 447400.00 | 24318000 |
+-----+-----+-----+-----+-----+
31 rows in set (0.000 sec)

MariaDB [(none)]> SELECT sum(SurfaceArea) AS "Total Surface Area", sum(Population) AS "Total Population"
-> FROM world.country
-> WHERE Continent = "North America";
-----+-----+-----+
| Total Surface Area | Total Population |
+-----+-----+-----+
| 24214470.00 | 482993000 |
+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [(none)]>
```

Respuestas Analíticas y Diagnóstico Técnico

- **¿Cuál es la ventaja de BETWEEN sobre >= y <=?** La principal ventaja es la **legibilidad y mantenimiento del código**. **BETWEEN** es un operador inclusivo que hace que la intención de la consulta sea clara para otros ingenieros, reduciendo el riesgo de errores lógicos al escribir múltiples condiciones **AND**.
- **Uso del Carácter Comodín %:** En el operador **LIKE**, el símbolo **%** representa "cualquier cantidad de caracteres (incluyendo cero)".
 - **%Europe%**: Busca "Europe" en cualquier parte de la cadena.
 - **Europe%**: Busca cadenas que *comiencen* con "Europe".
 - **%Europe**: Busca cadenas que *terminen* con "Europe".
- **Importancia de la función LOWER():** Aunque muchas configuraciones de MySQL son *case-insensitive* por defecto (dependiendo de la *collation*), en entornos de ingeniería de software profesional no se debe asumir esto. Utilizar **LOWER(columna)** garantiza que la búsqueda sea consistente independientemente de cómo se hayan ingresado los datos originalmente, evitando que registros como "CENTRAL" o "Central" queden fuera del reporte.

¡Felicidades! Has fortalecido tus habilidades de análisis de datos aplicando filtros condicionales y funciones de agregación en AWS.